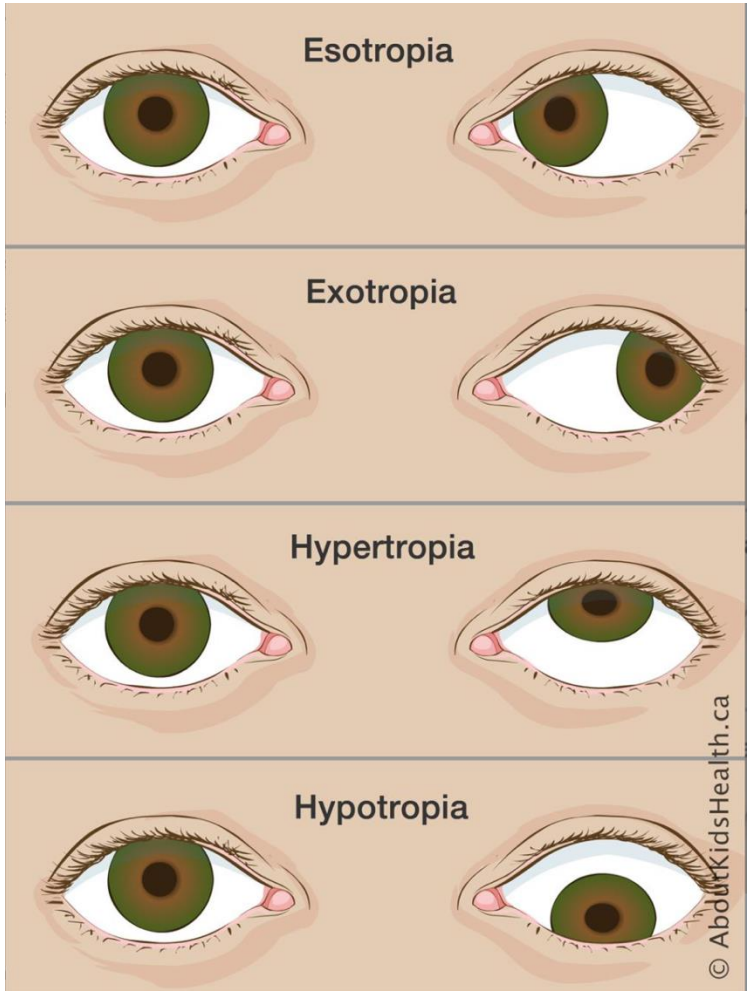
Virtual Reality Solution for Measuring Strabismus

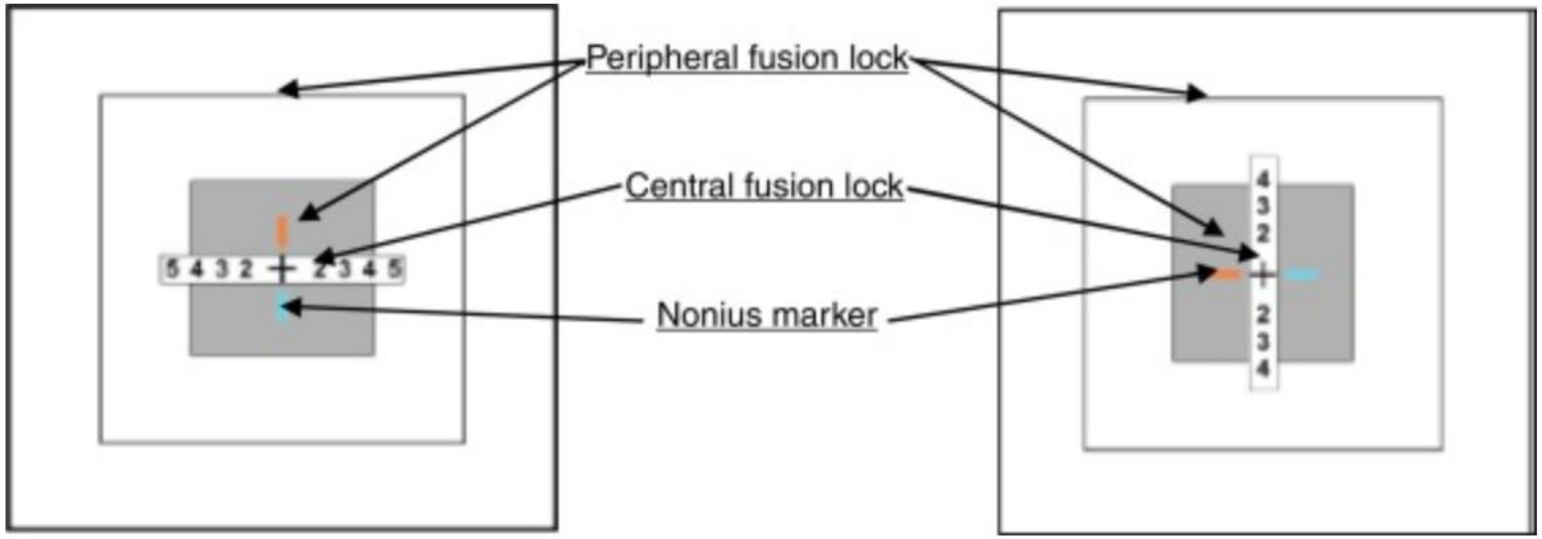
Joseph Tran, CS ‘22

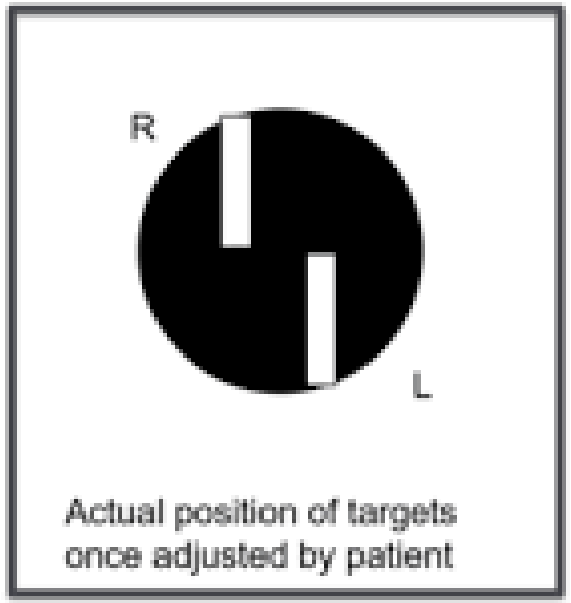
What is Strabismus?

- Put simply, “crossed eyes”
- Affects 4% of the population, ~13 million people
- Causes include weak eye muscles, cataract and nerve conditions
- Many questions about how this can affect perception of visual stimuli

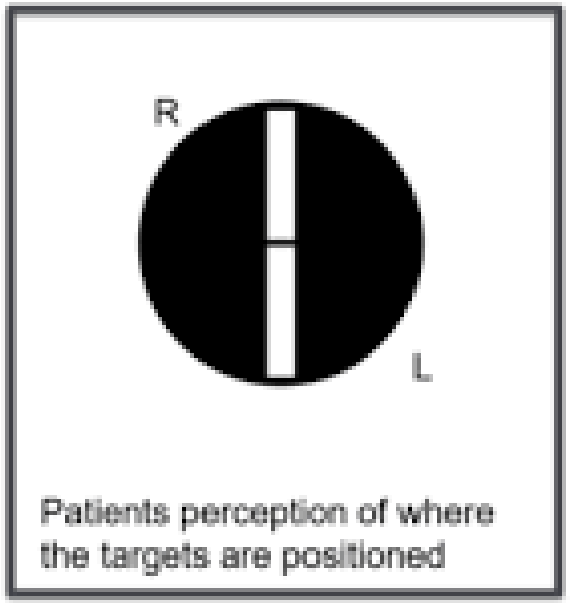


Disparometer Test





Actual position of targets once adjusted by patient

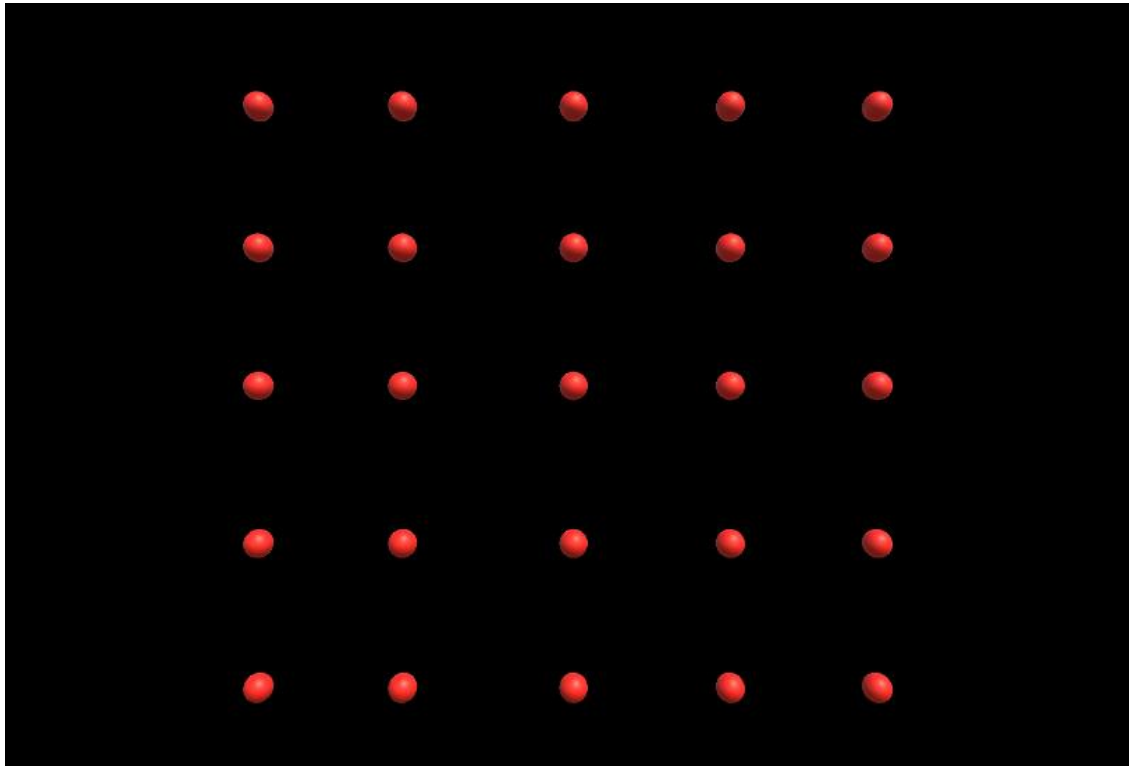


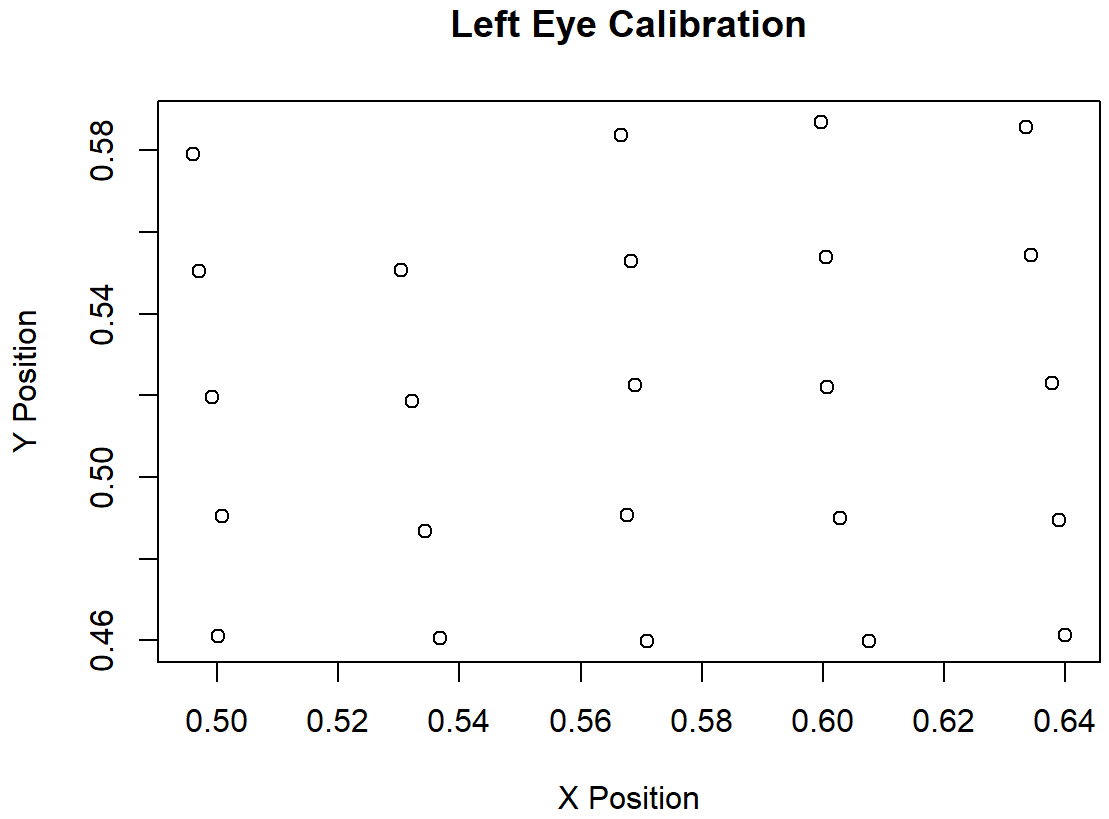
Patients perception of where the targets are positioned

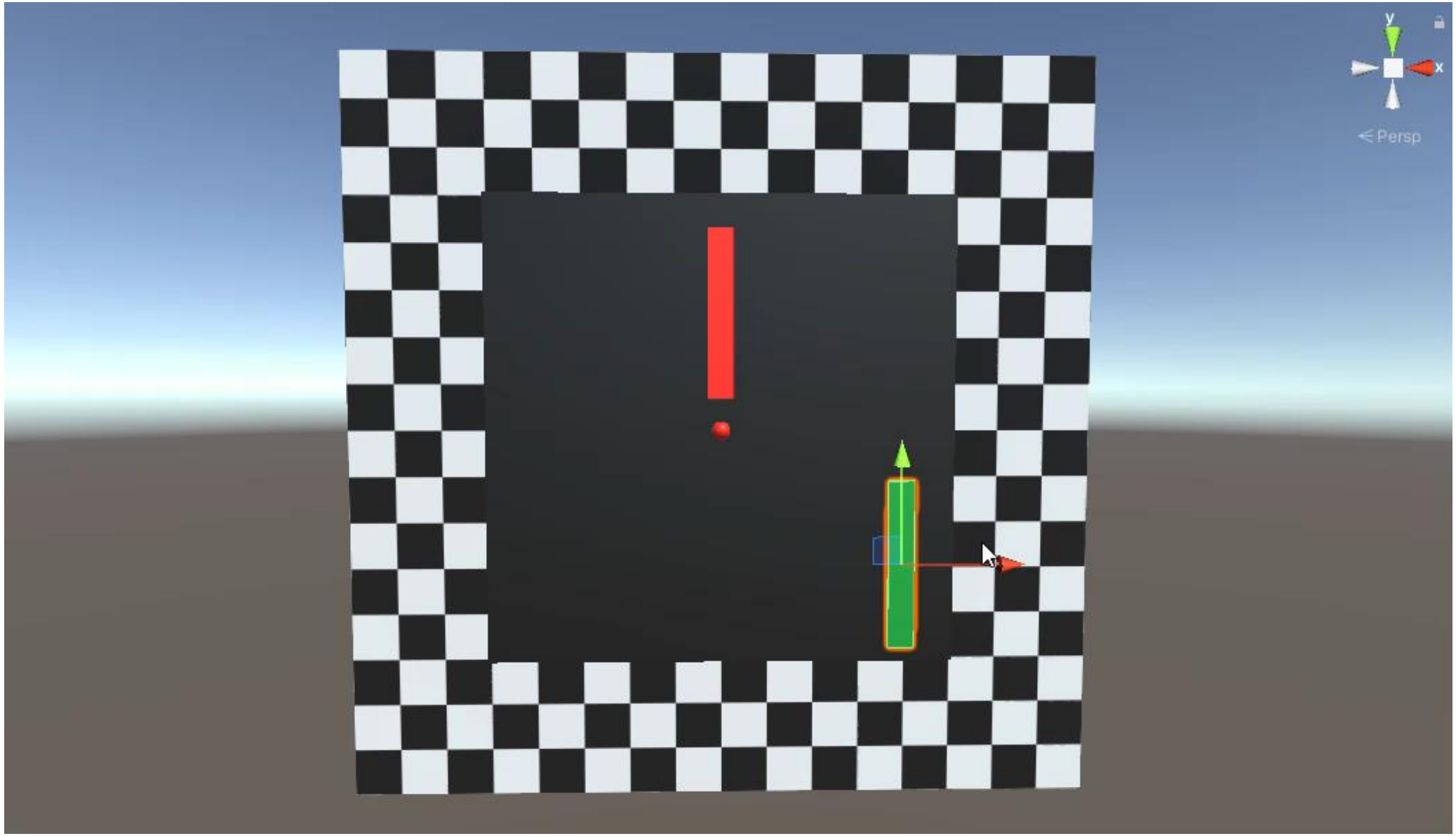
Research Project

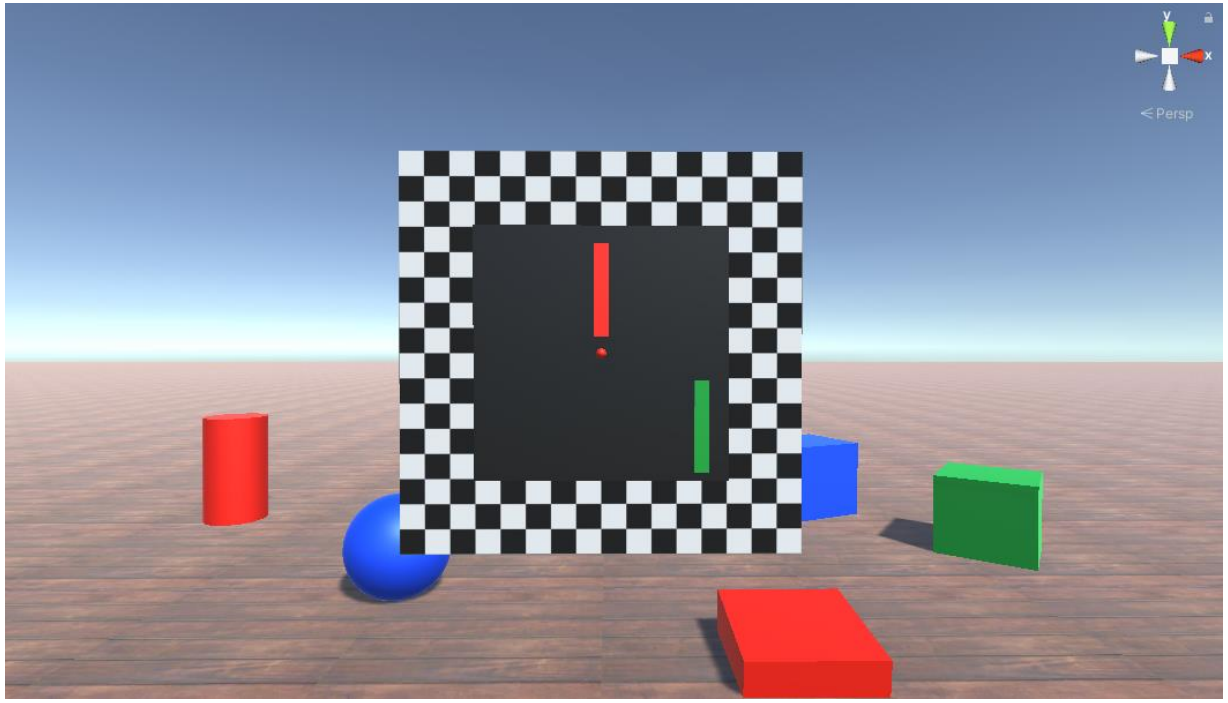
- Create a Disparometer test in virtual reality to see if modern VR headsets can be used as accurate measure of ocular deviation.
- Combine subjective and objective methods into single application
- Can we use VR to gain better insight to the effects of strabismus and how those affected perceive visual stimuli.

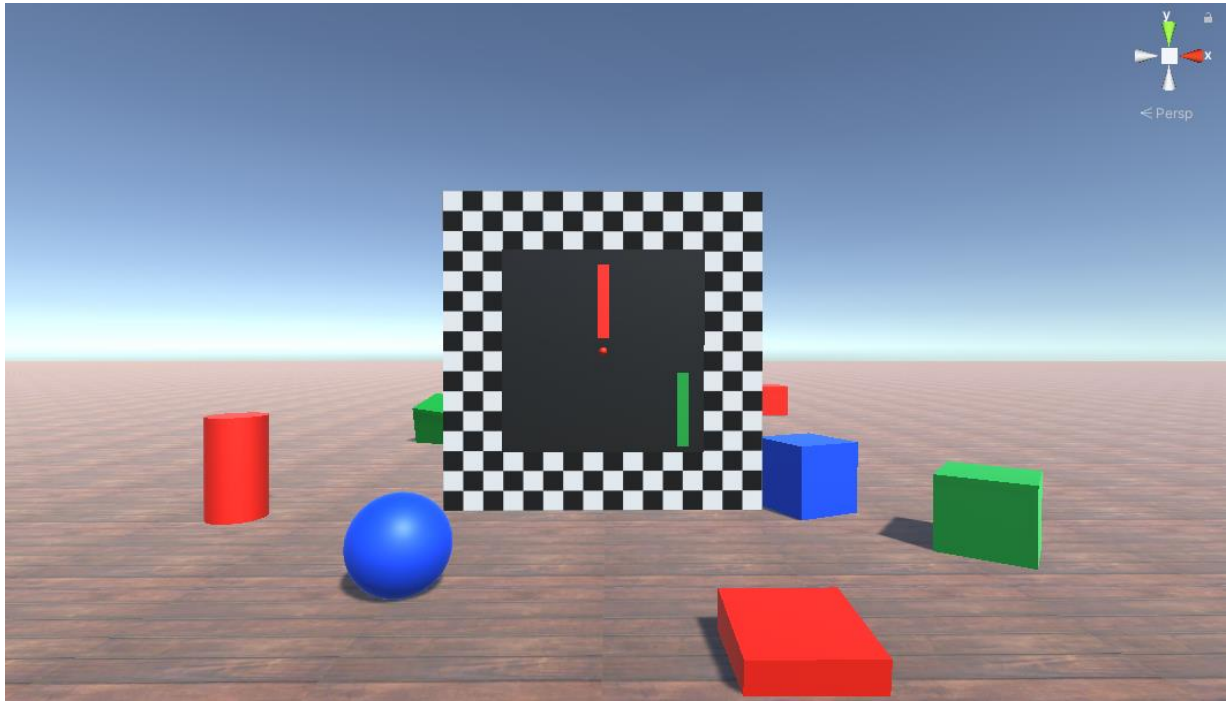
Initial Eye Calibration

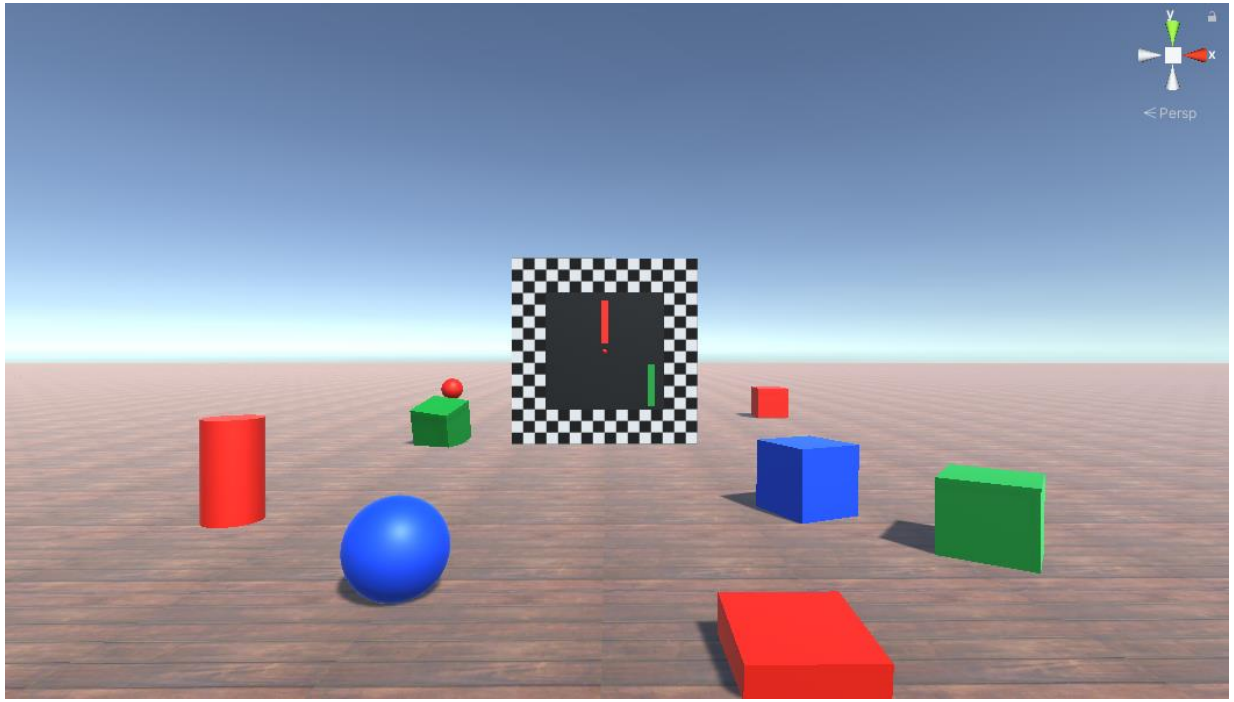


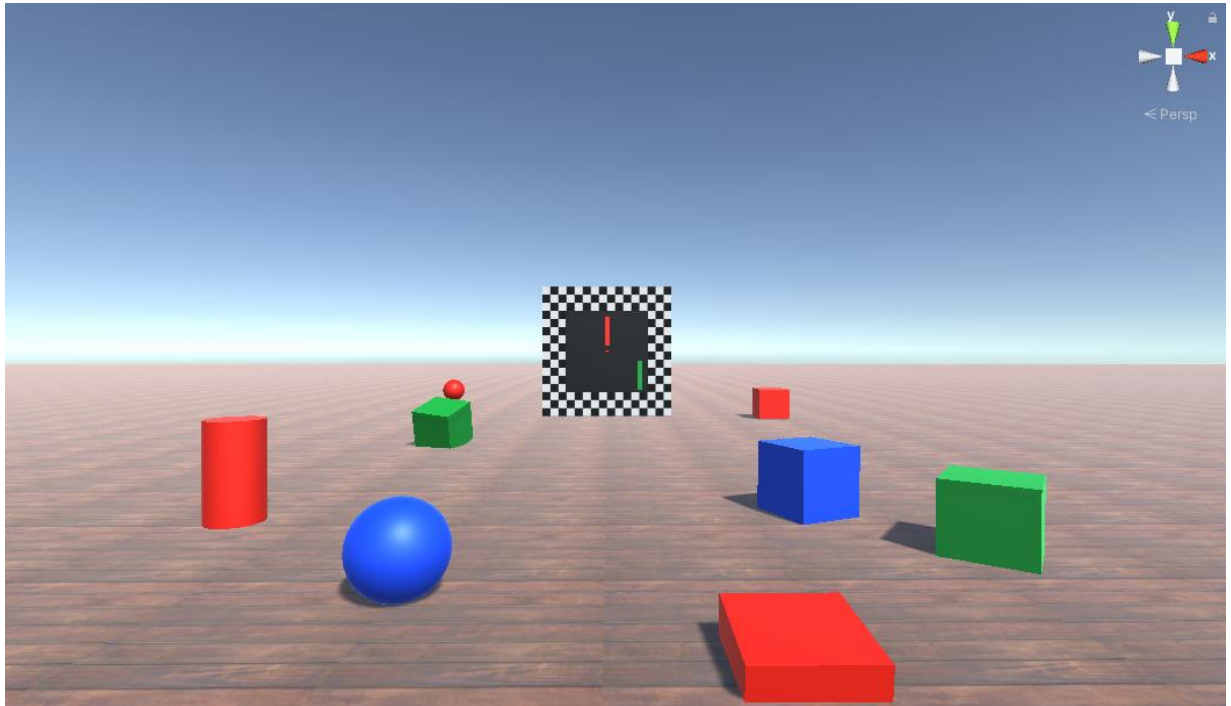


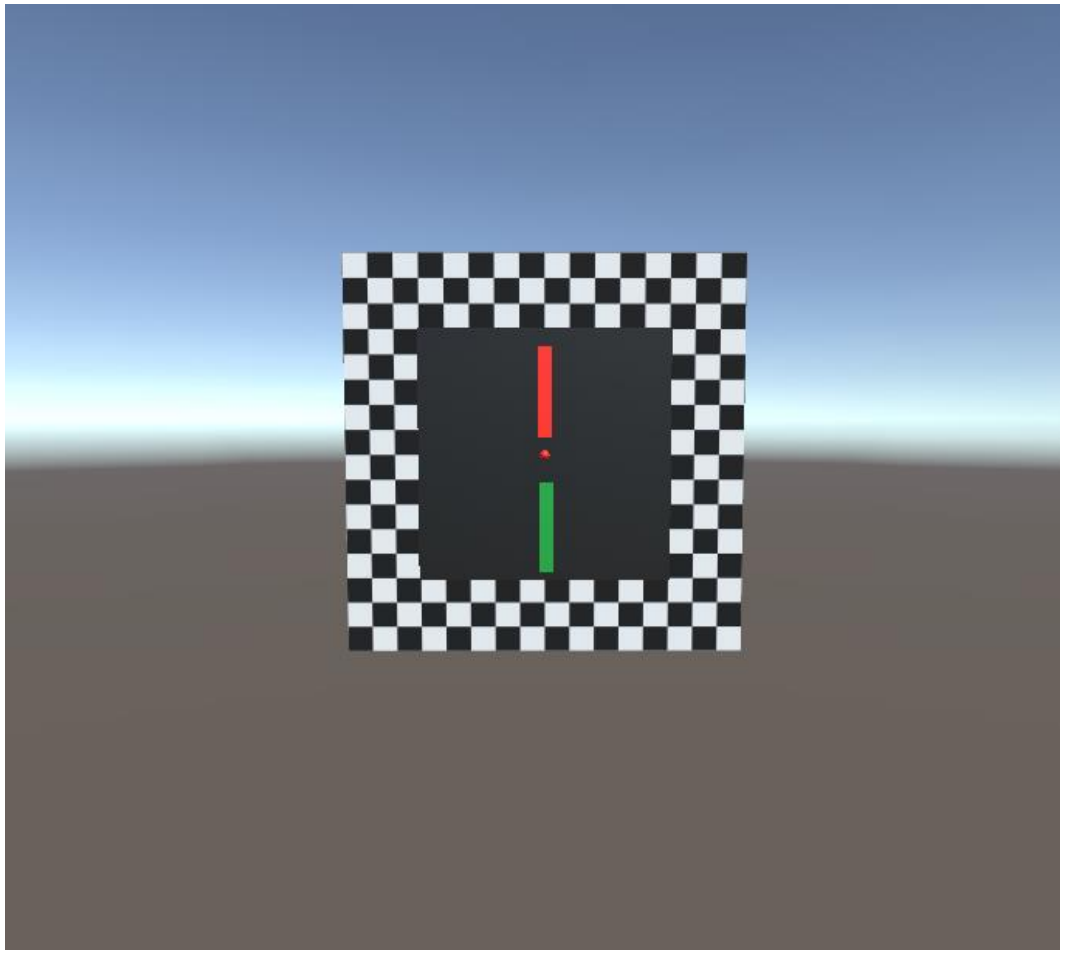

y
x
Persp

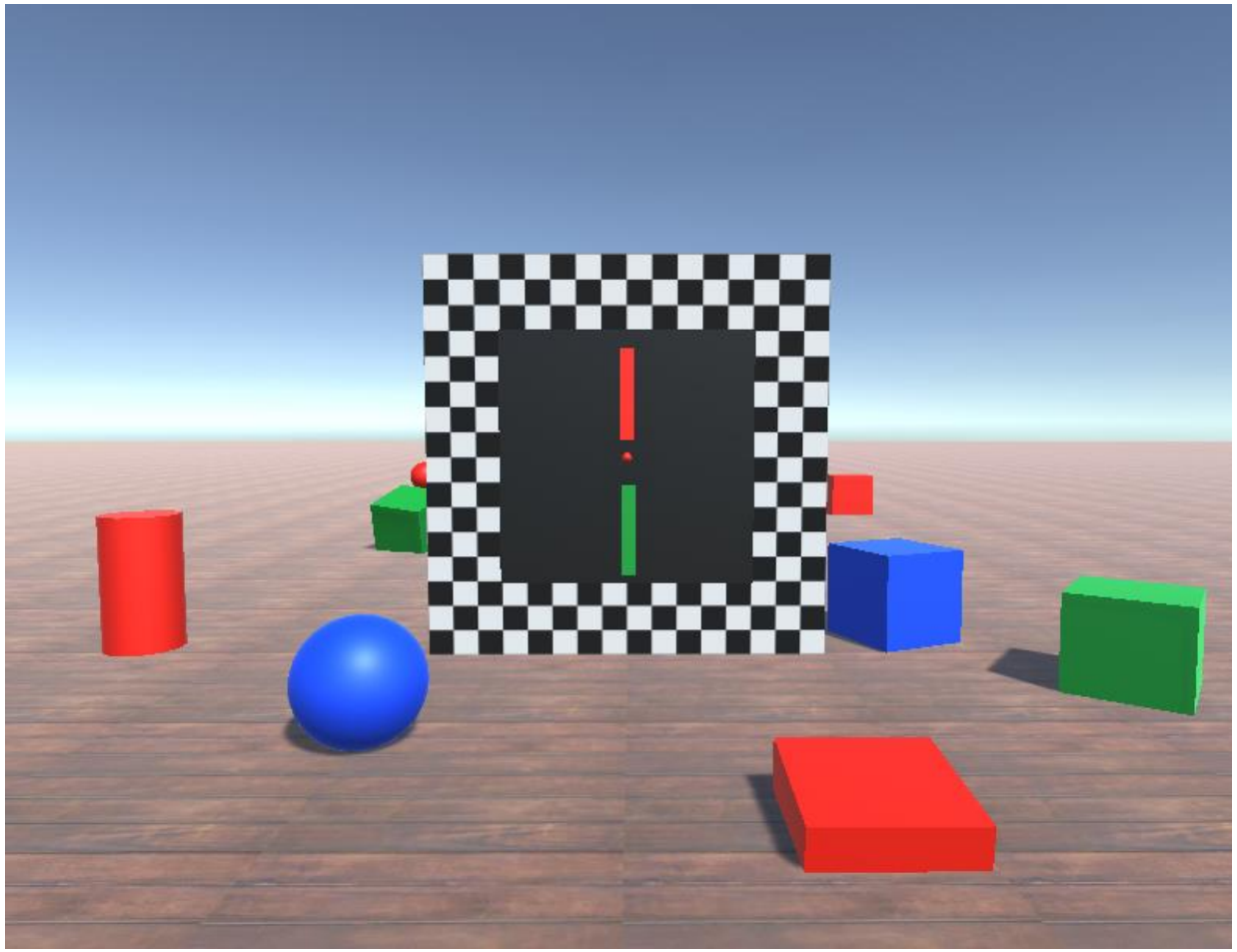

Persp

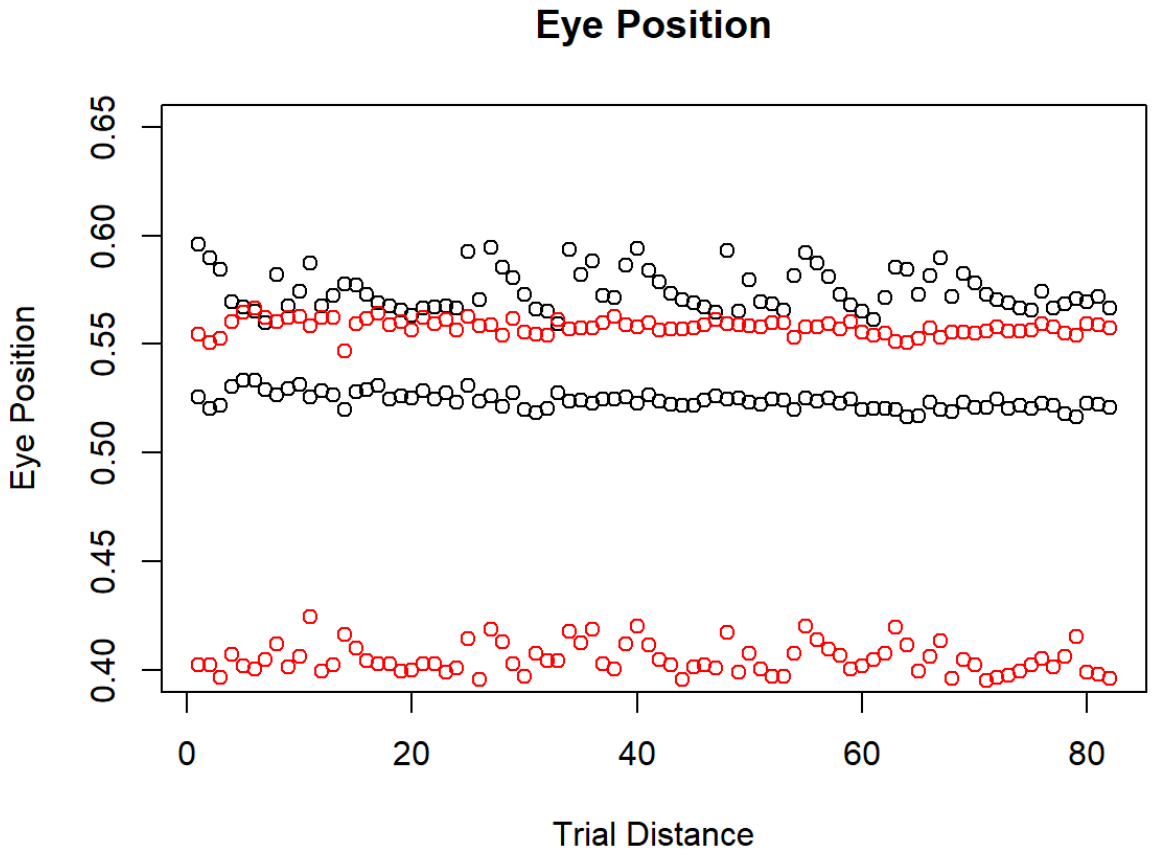

Persp

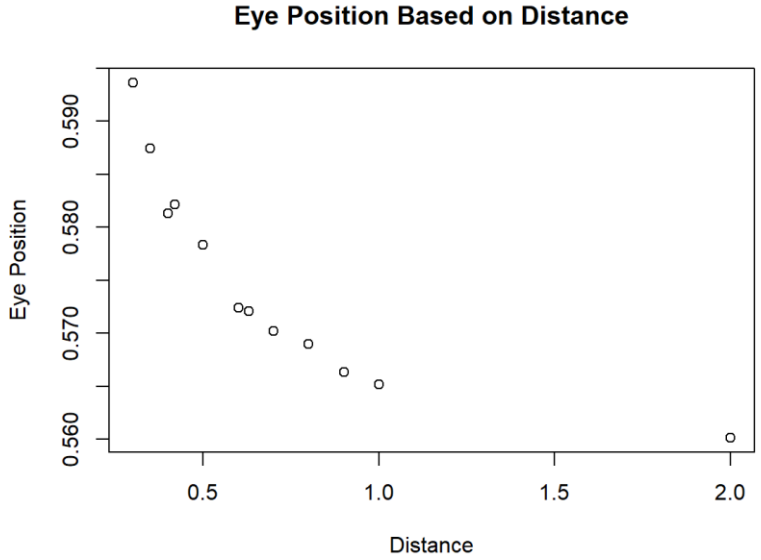

Persp

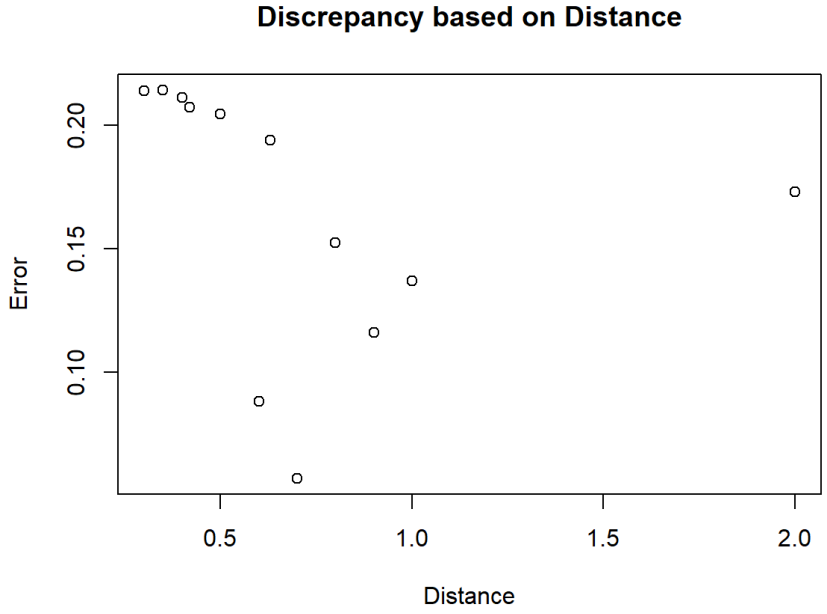

Persp






Eye Position
Eye Position
0.40
0.45
0.50
0.55
0.60
0.65
0
20
40
60
80
Trial Distance


Eye Position Based on Distance
Eye Position
0.560
0.570
0.580
0.590
0.5
1.0
1.5
2.0
Distance


Discrepancy based on Distance
Error
0.10
0.15
0.20
0.5
1.0
1.5
2.0
Distance